Peter Willendrup

Establishing the learning goals, a look at the programme

Learning goals:

1. Learn McStas basics

2. Build and operate simple instrument models, source + optics + sample + detector

3. Learn basics of instrument-optimisation and resolution calculations

5. Add Mantid / NeXus capabilities

6. Get a better idea of what you want to do with McStas, how to do it, how to get help

7. Get up-to-speed with latest developments and advanced features of McStas and related tools like NCrystal and McStasScript.

Enable your independent work with McStas

School programme

1 2 3

November 4th Beginners McStas	Time (GMT)	November 5th Instrument design	Time (GMT)	November 6th Advanced A	November 6th Adva
Welcome + setting learning goals 30 min McStas intro + general concepts 30 min McStas live demo / "Build-along" **Responsible: Peter**	9:00-10:30	30 min Polarisation 60 min Lesser known tips and tricks, including: - Tools for optimisation **Responsibles: Peter**	9:00-10:30	Advanced topics: NCrystal talk Using NCrystal in McStas & McStasscript **Responsible: Thomas**	Work on your own instrument w / help and guidance
Coffee Break	10:30-11:00	Coffee Break	10:30-11:00		Break
60min Guides and gravity: 20 min presentation 40 min practical **Responsible: Mads**	11:00-12:30	90 min McStasScript intro/exercise **Session lead: Mads**	11:00-12:30	Advanced topics interactive: Write your first component **Responsible: Peter**	Work on your own instrument w / help and guidance
Lunch break	12:30-13:30	Lunch break	12:30-13:30	Lunch break	Lunch break
Choppers / rotating optics: 20 min presentation 25 min practical 45 min Sample + Union overview talk **Responsibles: Mads + Peter**	13:30-15:00	McStas -> Mantid, NeXus: 20 min presentation 20 min demo 50 min exercise **Responsible: Peter**	13:30-15:00	Advanced talk/demo: McStas Union for detectors **Responsible: Mads**	Work on your own instrument w / help and guidance
Coffee Break	15:00-15:30	Coffee Break	15:00-15:30	Coffee Break	Break
45 min Sample exercise 45 min Union exercise **Responsibles: Peter + Mads**	15:30-17:00	Time of flight resolution, talk + exercise **Responsible: Mads**	15:30-17:00	Porting old comps and instruments to McStas 3 (and allow GPU execution) **Responsible: Peter**	Work on your own instrument w / help and guidance

Links to various infrastructure and services at:
https://isis2025.mcstas.org

School programme - day 1

November 4th Beginners McStas

Welcome + setting learning goals
30 min McStas intro + general concepts
30 min McStas live demo / "Build-along"

Responsible: Peter

Coffee Break

60min Guides and gravity:
20 min presentation
40 min practical

Responsible: Mads

Lunch break

Choppers / rotating optics:
20 min presentation
25 min practical
45 min Sample + Union overview talk
Responsibles: Mads + Peter

Coffee Break

45 min Sample exercise
45 min Union exercise

Responsibles: Peter + Mads

Intro lecture, general principles, sources + monitors,

Lectures + "recipe" exercises

Sample-lecture, including "advanced McStas grammar", sample exercises

In "cookbook" sections, think ahead toward your own project:

* Which neutron source
* What optics
* What sample

- K.I.S.S. for now

School programme - day 2

November 5th Instrument design
30 min Polarisation 60 min Lesser known tips and tricks, including: - Tools for optimisation **Responsibles: Peter**
Coffee Break
90 min McStasScript intro/exercise **Session lead: Mads**
Lunch break
McStas -> Mantid, NeXus: 20 min presentation 20 min demo 50 min exercise **Responsible: Peter**
Coffee Break
Time of flight resolution, talk + exercise **Responsible: Mads**

Lectures on polarisation and instrument optimisation technicals

ToF resolution, talk + exercise

Mantid-howto, lecture and demo

McStasScript intro/exercise

School programme - day 3, advanced stuff + work on own project

November 6th Advanced A	November 6th Advanced B
Advanced topics: NCrystal talk Using NCrystal in McStas & McStasscript **Responsible: Thomas**	Work on your own instrument w / help and guidance
	Break
Advanced topics interactive: Write your first component **Responsible: Peter**	Work on your own instrument w / help and guidance
Lunch break	Lunch break
Advanced talk/demo: McStas Union for detectors **Responsible: Mads**	Work on your own instrument w / help and guidance
Coffee Break	Break
Porting old comps and instruments to McStas 3 (and allow GPU execution) **Responsible: Peter**	Work on your own instrument w / help and guidance

For the exercise-based work-sessions

- You will benefit from working in pairs,   2 > 1

- Take turns being the "coder"  and the "parallel processor" 

Let’s get to it!